TiQE - Text to AI Generated Image Quality Evaluator

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Abstract. Recent improvements in generative models have greatly enhanced text-to-image synthesis, allowing for the generation of very realistic images from text descriptions. Still, it is difficult to assess the quality of output images because image quality is subjective and matching visual content to input text is complicated. Text-to-image generation involves complex cognitive processes, creating distinct challenges for evaluating image quality, which is often optimized for professional and user-generated content. Traditional image quality metrics primarily address natural distortions but fail to effectively capture the alignment between the generated image and its input prompt. This paper presents TIQE (Text-to-Ai Generated Image Quality Evaluator), a new approach to evaluate the quality of text-generated images. The proposed framework aims to leverage both perceptual quality metrics and semantic alignment techniques to provide a comprehensive evaluation of the generated images, assessing both their visual quality and their alignment with the given prompt.Code will be provided in the https://github.com/saikirankocherla/TiQE.git

1 Introduction

Text-to-image synthesis is currently a rapidly expanding field of artificial intelligence, driven by advances in deep learning models capable of generating high-quality images from text descriptions. This capability has tremendous potential in numerous applications, including digital art, content creation, and assistive technology. Despite these advances, evaluating the quality of synthesized images remains a challenging task. Traditional evaluation methods typically involve the use of subjective human judgments, being slow and not consistent. Metrics for automated evaluation, although objective, cannot be guaranteed to catch semantic coherence between the input text and the generated image.

Advancements in machine learning and computer vision have revolutionized multimedia communications, particularly with the rise of AI-Generated Content (AIGC). Generative models such as GANs, VAEs, LLMs, and diffusion models are now widely used to create realistic and artistic images, videos, and 3D scenes.However, visual quality assessment (IQA/VQA) for AI-generated and AI-enhanced content (AIGC/AIEC), emphasizing both the advancements and challenges in evaluating AI-generated media. Traditional quality assessment meth-

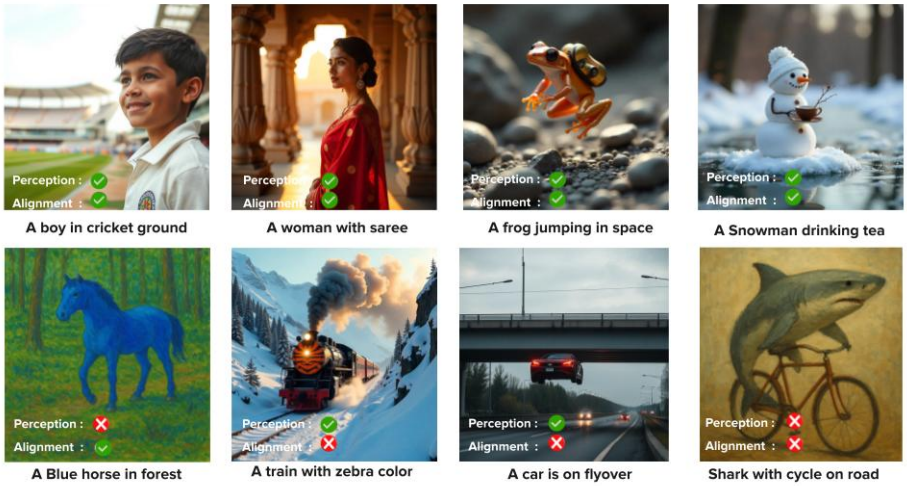


Fig. 1. Illustration of generated image cases showing all combinations of correct and incorrect perception and alignment.

ods, such as SSIM[23] and VMAF[24], struggle to accurately capture the distortions present in AI-generated images and videos. To address this, deep learning-based methods like LPIPS[25], DISTS[26], and CLIP-IQA[27] have been introduced, offering improved alignment with human perception. However, conventional AI-specific metrics, such as FID[29] and IS[28], often fail to fully capture the nuances of perceptual quality, leading to the development of more sophisticated benchmarks like PickScore[31], and VBench[30]. Subjective quality assessment, relying on human ratings such as Mean Opinion Scores (MOS), remains essential for evaluating AIGC. However, as AI-generated content evolves, existing methodologies face limitations in assessing aesthetics, naturalness, and the uncanny valley effect. Vision-language models (VLMs) offer promising solutions, but they require enhancements in abstraction levels to incorporate commonsense reasoning.

To overcome these limitations, we want to present TIQE (Text-to-AI Generated Image Quality Evaluator), a holistic system that can produce objective, consistent, and scalable quality estimates for text-to-image models.

2 Literature Survey

Conventional deep neural network (DNN)-based models for Image Quality Assessment (IQA) have predominantly been developed with the goal of evaluating quality impairments found in natural images. These models are typically trained to identify and quantify degradations such as Gaussian noise, motion blur, compression artifacts, and contrast distortions. However, such models are inherently

limited when applied to AI-generated images (AGIs), particularly those synthesized through prompt-based text-to-image models. This discrepancy arises due to the fundamentally different distribution and semantics in AI-generated images compared to real-world photographs. While natural image IQA focuses on pixel-level artifacts and structural distortions, AGIs often exhibit issues like prompt misalignment and semantic inconsistencies that lie beyond traditional distortion-based modeling.

Efforts have been made to adapt traditional IQA models to this emerging domain by retraining or fine-tuning them on AGI datasets. While this approach may yield superficially improved IQA performance scores, it inadvertently introduces bias: the model learns to recognize statistical patterns typical of AI-generated content rather than genuinely assessing quality in terms of human perceptual alignment or semantic fidelity. Consequently, these retrained models may deliver inflated scores that do not correspond to human judgments of quality, naturalness, or realism, especially in cases where prompt-image correspondence is critical.Classical no-reference IQA methods such as BRISQUE [14], ILNIQE [15], and NIQE [16] are grounded in Natural Scene Statistics (NSS), where statistical regularities in natural images are used to infer quality without a reference image. Although computationally efficient and interpretable, these models struggle to generalize to AI-generated images due to the latter's synthetic nature and lack of natural image statistics. Consequently, these handcrafted models often fail to recognize perceptual issues in AGIs, particularly those involving semantic dissonance or stylistic aberrations.

Deep learning-based IQA models such as DBCNN [17] and HyperIQA [18] have been introduced to overcome the limitations of traditional models by leveraging Convolutional Neural Networks (CNNs) to extract hierarchical features. These models show improved robustness when handling both synthetic and authentic distortions by learning spatial and structural patterns directly from data. However, their reliance on low-level visual cues still constrains their ability to capture the complex semantics and prompt fidelity required for evaluating AGIs.To address the shortcomings of CNNs in modeling long-range dependencies and contextual information, Vision Transformer (ViT)-based IQA models have emerged. Notable examples include IQT [19], MUSIQ [20], TReS [32], and MANIQA [33]. These models employ self-attention mechanisms and multi-scale encoding to integrate both local and global information, enabling better modeling of perceptual quality across various resolutions and distortions. While they offer promising results for natural and distorted images, their performance still diminishes when applied to AI-generated images where textual alignment and semantic integrity play a pivotal role. The limitations of existing IQA models in handling AGIs have led to the emergence of Large Multi-Modality Models (LMMs) that incorporate language understanding and visual perception for more comprehensive assessment. These models leverage the synergy between vision and language to better understand prompt-image relationships, which is

especially vital for evaluating Text-to-Image (T2I) generated content.

A prominent example of this paradigm shift is the MA-AGIQA framework [2], which integrates MANIQA [33] as the deep learning backbone for visual quality encoding, and mPLUG-Owl2 [3] to extract semantic cues from text prompts. This combination is further enhanced using a Mixture of Experts (MoE) mechanism [34], allowing adaptive fusion of both perceptual and semantic features. The MoE architecture enables the model to dynamically weigh the contributions of image quality and prompt alignment cues, thereby achieving a more nuanced and task-aware quality prediction.Further extending the multimodal assessment paradigm, IPCE(TSP-MGS) [4] introduces an innovative approach that utilizes the CLIP encoder [5] to align visual and textual embeddings. In their method, the degree of similarity between the AI-generated image and the guiding prompt is measured through carefully crafted text templates, representing varying levels of semantic similarity. These scores are subsequently refined using regression-based modeling to deliver continuous and interpretable quality scores. This design effectively bridges the semantic gap between text and image representations, enabling better alignment-based evaluation.

Drawing inspiration from IPCE, the current research adopts and advances their methodology by incorporating task-specific prompt design and multigranularity similarity analysis. In our approach, coarse-grained templates are developed to evaluate two distinct facets of image quality i.e, semantic alignment and visual perception. In the proposed framework, alignment prompts are formulated to quantitatively evaluate the correspondence between the generated image and its conditioning textual description, whereas perception prompts are designed to assess intrinsic image attributes such as clarity, structural coherence, and photorealism. To operationalize this dual evaluation, the architecture incorporates two task-specific output heads, each associated with its respective supervision signal. The alignment head is trained against ground-truth alignment scores, while the perception head is trained against ground-truth perception scores. Task-specific losses are computed independently and aggregated during optimization, ensuring that the model learns both objectives without entanglement.

To further increase fidelity, we introduce fine-grained analysis by computing word-level similarities between the initial prompt and both global image embeddings and local patch-wise features. This allows the model to capture nuanced distortions and omissions that may not be evident at the sentence level. Such granularity provides a more detailed and discriminative representation of quality, essential for complex generative failures. Finally, to ensure fair comparison and reproducibility, our work adheres to the same dataset split protocol as adopted in IPCE [4]. By maintaining consistent training and testing procedures across the AGIQA-1K and AGIQA-3K benchmarks, we ensure that performance gains are attributable to methodological improvements rather than experimental variance.

3 Related Work

3.1 General-Purpose Image Quality Assessment

Conventional Image Quality Assessment (IQA) approaches have been primarily developed to measure distortions in natural or synthetically degraded images. Traditional full-reference metrics such as Peak Signal-to-Noise Ratio (PSNR) [22] and Structural Similarity Index Measure (SSIM) [23] estimate image fidelity based on pixel-wise comparisons with a pristine reference. These metrics are mathematically convenient but have limited correlation with human perceptual assessments, especially under complex or semantic distortions.

To address this, no-reference methods such as BRISQUE [14] analyze statistical deviations in natural scene statistics (NSS) without requiring reference images. These handcrafted feature-based models detect low-level artifacts such as blur, noise, and compression distortions effectively in real-world scenarios. However, they inherently lack the capability to assess higher-level semantics or contextual correctness, which are critical when dealing with AI-generated images (AGIs) that may appear visually sharp yet semantically incoherent.Deep learning-based models have further pushed the boundaries of distortion modeling. For instance, WaDIQaM [21] employs a Siamese network to learn visual similarity, improving robustness to unseen distortions. HyperIQA [4] uses a hypernetwork to dynamically adjust feature representations for blind IQA tasks. While these models generalize well to authentic and synthetic degradations, they are largely text-agnostic they do not account for semantic correctness or prompt-image alignment, which are essential for assessing AGIs.

3.2 Prompt-Aware and Vision-Language IQA Models

Prompt-based generative models have motivated IQA methods emphasizing vision-language alignment. Vision-Language Models (VLMs) such as CLIP [5] capture semantic relations between text and images, enabling prompt-aware IQA approaches.CLIPScore and ImageReward [35] compute alignment via cosine similarity between global embeddings of images and prompts, but are sensitive to phrasing and coarse in granularity. Peng et al. [4] improved this with patch-level embeddings and prompt variants, yet fixed feature representations limit adaptability.Wang et al. [2] addressed these issues by combining a deep visual IQA backbone (MANIQA [33]) with a frozen LMM (mPLUG-Owl2 [3]), fusing perceptual and semantic features. While effective, reliance on frozen LMMs still restricts adaptation to diverse prompts.

3.3 Multimodal and Prompt-Conditioned AGIQA

Backbone choice strongly affects multimodal IQA. ViT-based CLIP models (e.g., ViT-B/32) provide stronger contextualization than CNNs like RN50. Works such as BLIP-2 [6] and ALIGN [7] show that combining frozen visual encoders with

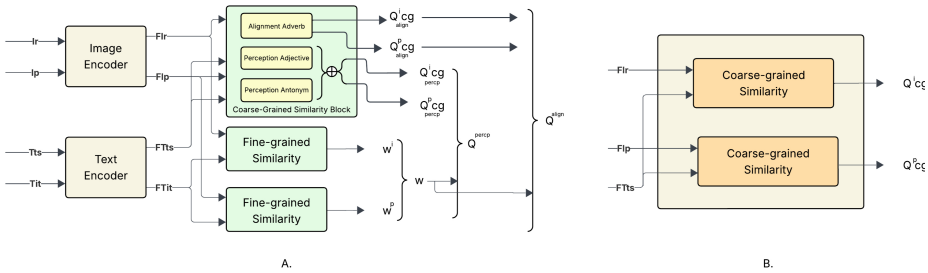


Fig. 2. A)Model Architecture of TiQE and B)Coarse-grained similarity Block.

trainable language models enhances cross-modal alignment, benefiting scalability and semantic precision.Beyond architecture, adaptive similarity and prompt refinement are gaining traction. DINOv2 [8] demonstrates the semantic power of self-supervised ViTs without text, while PromptAlign [9] refines prompts dynamically at inference. ImageBind [10] further extends alignment to multiple modalities, including audio.

Datasets such as AGIQA-1K and AGIQA-3K support rigorous evaluation, with methods like Prompt-AIGIQA [12] and PK-AIGIQA-4K [11] achieving state-of-the-art results. These approaches combine human-annotated datasets, prompt conditioning, and CLIP-based encoders to deliver perceptually aligned predictions, capturing fine-grained quality variations across styles. While VLMs form a strong foundation, current AGIQA methods still trade off generalization, precision, and adaptability. Progress will require balancing self-supervised encoders, dynamic prompt encoding, and hierarchical similarity modeling.

4 Methodology

4.1 Image-Processing and Embeddings

Initially, we acquired the archive of the AGIQA-3k[13] data set, which contains N images. And resized all images to (3,224,224).We next obtained CLIP image-encoder embeddings for each full image yielding a matrix $F_I^{(r)} \in R^{N\times 512}$ —for Coarse-Grained similarity and fine-grained similarity where we used coarse-grained similarity for alignment and preception and fine-grained for detailed quality understading which discussed more on respective sections. Also,each image was partitioned into 64 non-overlapping patches; the same CLIP model independently encoded these patches to produce $F_I^{(p)} \in R^{(N\times 64)\times 512}$. we used $F_I^{(p)}$ for fine-grained similarity. Since all 64 patches originate from the same source image, we then averaged each group of 64 patch embeddings to reconstruct an image-level descriptor $\bar{F}_I^{(p)} \in R^{N\times 512}$. we used for $\bar{F}_I^{(p)}$ coarse-grain similarity. Finally, we encapsulated this workflow in a single function that returns:

1. the resized-image embeddings $F_I^{(r)}$,

2. the raw patch embeddings $F_I^{(p)}$,
3. the averaged patch embeddings $\bar{F}_I^{(p)}$.

Table 1. Summary of image embeddings

Embedding Set	Shape	Description
$F_I^{(r)}$	$(N, 512)$	Full-image embeddings
$F_I^{(p)}$	$(N \times 64, 512)$	All 64 patch embeddings
$\bar{F}_I^{(p)}$	$(N, 512)$	Per-image averaged patches

4.2 Text Prompt Construction and Embeddings

To enable multi-granularity evaluation of AI-generated images, we construct two categories of text prompts for each image. The first category comprises task-specific prompts (T_{ts}), which are designed to assess both perceptual quality and alignment fidelity. For each initial prompt pt, we generate 12 textual variants using three linguistic strategies:

(i) adverb-based: $T_{ts}^{\text{adv}} =$ "A photo that {adv} matches {pt}",
where {adv} $\in$ {*badly*, *poorly*, *fairly*, *well*, *perfectly*};

(ii) adjective-based: $T_{ts}^{\text{adj}} =$ "A photo of {adj} quality",
where {adj} $\in$ {*bad*, *average*, *good*};

(iii) antonym-based: $T_{ts}^{\text{ant}} =$ "{ant} photo",
where {ant} is an antonymic modifier (*bad*, *good*, *perfect*).

Each T_{ts} is encoded using the CLIP text encoder to obtain a sentence-level embedding for coarse-grained similarity measurement.

The second category consists of the initial prompts (pt), which are the textual descriptions originally used to generate the images. Each prompt is tokenized into individual words indexed as FW_k, where $k \in \{1, 2, \ldots, K\}$. Word embeddings FW_k are extracted using the CLIP ViT-B/32 encoder, and all sequences are padded or truncated to a fixed length of 16, yielding a tensor of shape $[N, 16, 512]$ for N prompts. These are used for fine-grained token-level similarity evaluation.

This two-tiered embedding strategy—sentence-level via T_{ts} and token-level via FW_k—enables a comprehensive evaluation of both perceptual and semantic alignment in AIGIs.

4.3 Coarse-Grained Quality

The coarse-grained similarity measurement provides a fundamental framework for assessing AI-generated image quality by establishing semantic alignment between visual content and textual quality descriptions. This approach addresses two critical challenges in quality assessment: (1) the need for holistic quality evaluation beyond pixel-level distortions, and (2) the requirement for interpretable quality metrics that correlate with human perception.

Image-level similarity measures global alignment between the resized image and quality prompts, capturing overall composition and style adherence:

$$S_j^I = \frac{F_{I_r} \cdot F_{T_j}}{\|F_{I_r}\| \|F_{T_j}\|}, \quad j \in \{1, \ldots, L\} \tag{1}$$

Patch-level similarity assesses local quality through averaged patch features, enabling detection of fine-grained artifacts that might be overlooked in global analysis:

$$S_j^P = \frac{\bar{F}P \cdot FT_j}{\|\bar{F}P\| \|FT_j\|} \tag{2}$$

The dual-level similarity computation stems from the observation that human evaluators naturally assess images through both global impressions and local scrutiny. While traditional methods often focus exclusively on low-level features, this approach mirrors the hierarchical nature of human visual perception, where both macroscopic composition and microscopic details contribute to quality judgments.

Quality probability conversion transforms raw similarity scores into probabilistic distributions using softmax computation, accounting for the inherent subjectivity in quality assessment:

$$p_j^I = \frac{\exp(S_j^I)}{\sum_{l=1}^{L} \exp(S_l^I)}, \quad p_j^P = \frac{\exp(S_j^P)}{\sum_{l=1}^{L} \exp(S_l^P)} \tag{3}$$

This probabilistic framework was motivated by psychovisual studies showing that human quality ratings often cluster around discrete levels with smooth transitions between them. The softmax operation effectively models this behavior while maintaining ordinal relationships between quality grades.

Quality score regression converts probabilistic outputs into continuous scores suitable for regression tasks, preserving the interpretability of discrete levels while enabling fine-grained quality comparisons:

$$Q_{cg}^* = \frac{L}{L-1} \left(\sum_{j=1}^{L} j \cdot p_j^* - 1 \right), \quad * \in \{I, P\} \tag{4}$$

Final score integration implements an adaptive weighting mechanism that automatically balances global and local quality aspects based on their relative importance for each image:

$$Q_{cg} = \alpha Q_{cg}^I + (1 - \alpha) Q_{cg}^P \tag{5}$$

The dynamic weighting reflects the empirical finding that different image types require different evaluation emphases, for instance, architectural renderings may demand stricter local quality control than impressionistic artworks. This flexibility represents a significant advancement over fixed-weight combination approaches prevalent in traditional quality assessment methods.we can make this

trainable or set to 0 which considers only patches contribution or set to 1 which considers only resized original image contribution for score calculation.
In this manner this is applied to all the 2982 images on our dataset to get (2982,) tensor representing coarse grained quality scores for all the entries in the AGIQA-3k Dataset

4.4 Fine-Grained Similarity Measurement

While coarse-grained similarity measures the overall alignment between an image and its associated sentence-level prompt, fine-grained similarity captures more detailed token-wise correspondence between the image content and the keywords within the initial prompt.

Let the initial prompt be tokenized into K individual words, each represented by its CLIP embedding as FW_k, where $k \in \{1, 2, \ldots, K\}$. Given an image I_r, its global CLIP feature representation is denoted as F_{I_r}. The fine-grained similarity between the image and each word in the prompt is computed using the cosine similarity between F_{I_r} and FW_k. The aggregated image-level similarity score w_I is then obtained by averaging these word-level similarities:

$$w_I = \frac{1}{K} \sum_{k=1}^{K} \frac{F_{I_r} \cdot FW_k}{\|F_{I_r}\| \cdot \|FW_k\|}. \tag{6}$$

Similarly, we compute the patch-level similarity between the patch feature F_{P_n} and each word embedding FW_k. Let F_{P_n} denote the feature embedding of the n-th patch P_n within an image. For each patch, we calculate the mean cosine similarity with all K words in the initial prompt and take the maximum response across all patches to obtain the patch-level fine-grained score w_P:

$$w_P = \max_{n \in \{1,\ldots,N\}} \left(\frac{1}{K} \sum_{k=1}^{K} \frac{F_{P_n} \cdot FW_k}{\|F_{P_n}\| \cdot \|FW_k\|} \right). \tag{7}$$

Together, w_I and w_P capture both global and local semantic consistency between the generated image and the word-level components of its initial prompt. These fine-grained similarity scores are then fused with the coarse-grained similarity measures to provide a holistic assessment of AI-generated image quality.

We convert the fine-grained similarity scores into a quality score Q_{fg} by averaging the image-level (w_I) and patch-level (w_P) similarities, scaled by a factor L:

$$Q_{fg} = \frac{w_I + w_P}{2} \times L. \tag{6}$$

The final predicted quality score Q for an AIGI is obtained by combining both coarse-grained and fine-grained components. Specifically, we use a weighted sum of the coarse-grained scores from the image and patch branches, Q_{cg}^I and Q_{cg}^P, respectively, along with the fine-grained score Q_{fg}:

$$Q = \alpha \cdot Q_{cg}^{I} + (1 - \alpha) \cdot Q_{cg}^{P} + Q_{fg}, \tag{7}$$

where $\alpha \in [0, 1]$ is a hyperparameter used to balance the contribution of the global image and patch-level coarse-grained components.

To train the model, we use the Mean Absolute Error (MAE) loss between the predicted quality Q and the ground-truth subjective quality score Q':

$$\mathcal{L} = |Q - Q'|. \tag{8}$$

The model incorporates two specialized output heads i.e, a perception head for object/entity recognition and an alignment head for contextual correctness evaluation. During training, the predictions of each head are independently compared with their respective ground truths, and separate loss functions are computed(Q^{percep} and Q^{align}) as shown in Figure 2. These losses are then combined to jointly optimize the model for both perceptual accuracy and contextual alignment, enabling robust performance across scenarios where either or both aspects may succeed or fail.

5 Experiements

5.1 Implemtation Details

All experiments are performed on a PC equipped with an NVIDIA A40 GPU, using PyTorch 1.12.0 and CUDA 12.4. We load the ViT-B/32 as the backbone of our method based on experimentation (Fig 4, where input images are resized to 224×224. The model is optimized using the AdamW optimizer with a learning rate of 5×10^{-6} and a weight decay of 5×10^{-4}. A cosine annealing learning rate scheduler is applied to gradually reduce the learning rate every 5 epochs. The batch size is set to 16.To ensure reproducibility and fairness in comparison we dividing AGIQA-3k dataset into training and testing sets in a 4:1 ratio.

5.2 Hyperparameters

- **Backbone Model:** ViT-B/32 (Vision Transformer)
- **Input Image Size:** 224×224
- **Optimizer:** AdamW
- **Learning Rate:** 5×10^{-6}
- **Weight Decay:** 5×10^{-4}
- **Learning Rate Scheduler:** Cosine annealing (updated every 5 epochs)
- **Batch Size:** 16
- **Dataset Split:** 80% training / 20% testing (4:1 ratio)

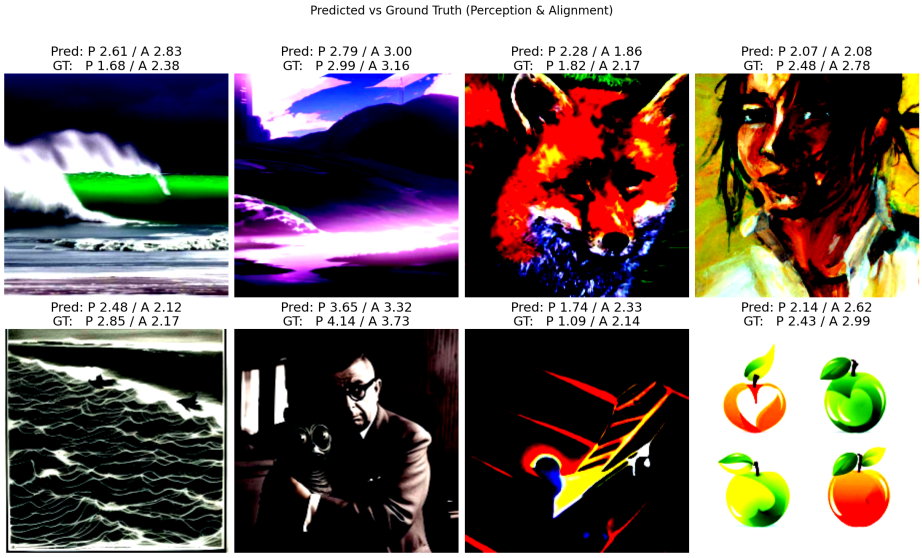


Fig. 3. Illustration of some AIGIs with subjective score and prediction for Perception and Alignment

6 Results and Discussion

The analysis of CLIP-based similarity scores for prompt-image alignment using both ViT-B/32 and RN50 models revealed meaningful insights into how different backbones encode semantic alignment.As shown in Figure 4, the ViT-B/32 model yields higher and more dispersed CLIP similarity scores than RN50, indicating that its transformer-based architecture captures richer semantic associations. In contrast, RN50 produces a narrower distribution, reflecting more conservative estimations.

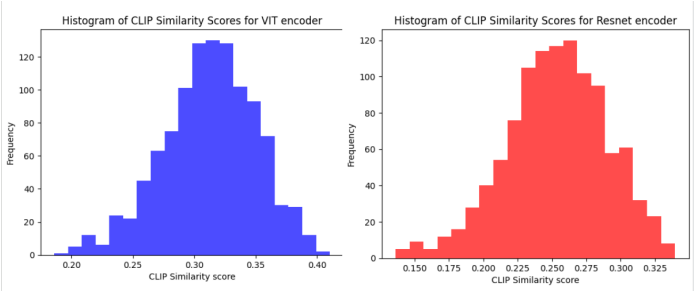


Fig. 4. Distribution of prompt-image CLIP similarity scores. Left: ViT-B/32. Right: RN50.

The results presented in Tables 2 and 3 for a comparative evaluation of the proposed method (Ours, denoted as TiQE) against three recent SOTA baselines till date, PK-AIGIQA [11], Prompt-AIGIQA [12] and TSP-MGS [4] on

the AGIQA-3k dataset. In terms of image perception (Table 2), TiQE outperforms both baselines, achieving the highest SRCC (0.8947), indicating superior consistency and accuracy in quality prediction. Similarly, for text-image alignment assessment (Table 3), TiQE again shows marked improvements with an SRCC of 0.7766, surpassing the existing methods by a significant margin. These results demonstrate the effectiveness of TiQE in both perceptual quality estimation and multimodal alignment. Figure 3 presents a qualitative comparison

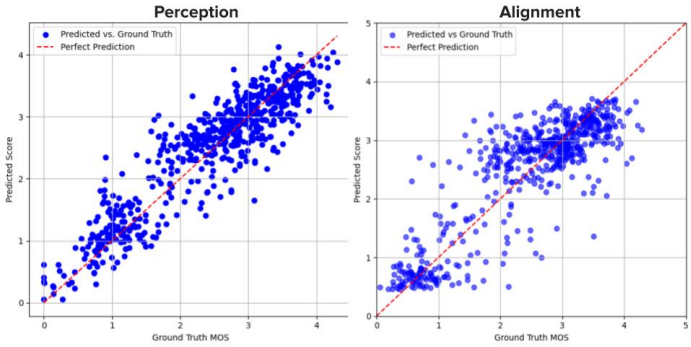


Fig. 5. Predicted vs. ground truth MOS scores for Perception and Alignment

between predicted and ground truth (GT) scores for both perception (P) and alignment (A) across diverse visual samples. Overall, the model demonstrates a strong ability to capture relative trends in both metrics, with most predictions lying within a ± 0.5 deviation from the corresponding ground truth values. This suggests that the dual-headed architecture is effective in learning to disentangle perceptual quality from semantic alignment. The dual-headed architecture demonstrates consistent correlation with ground truth, with most deviations under ± 0.5 (Figure 5), confirming effective joint learning of visual quality and semantic alignment. Notable errors occur in highly stylized, abstract, or color-distorted images, where perception scores are sometimes inflated due to visual saliency and alignment scores are underestimated when semantic cues are subtle or occluded. These results indicate that while the model generalizes well to typical scenes, further robustness is required for out-of-distribution visual styles and semantically ambiguous content.

Table 2. Comparison of SRCC and PLCC for Image Perception on AGIQA-3k dataset.

Method	SRCC	PLCC
PK-AIGIQA [11]	0.8249	0.8773
Prompt-AIGIQA [12]	0.8154	0.8807
TSP-MGS [4]	0.8901	**0.927**
Ours(TiQE)	**0.8948**	0.9236

Table 3. Comparison of SRCC and PLCC for Text-Image Alignment on AGIQA-3k dataset.

Method	SRCC	PLCC
PK-AIGIQA [11]	0.6624	0.7801
Prompt-AIGIQA [12]	0.608	0.7846
TSP-MGS [4]	0.7734	**0.8773**
Ours(TiQE)	**0.7766**	0.8552

7 Conclusion and Future Work

In this work, we proposed TiQE, a novel model for image quality estimation and text-image alignment, which outperformed existing methods like PK-AIGIQA, Prompt-AIGIQA and TSP-MGS on the AGIQA-3k dataset. TiQE achieved the highest SRCC scores, demonstrating its effectiveness in predicting human-judged quality scores and capturing semantic relationships between images and prompts in both perception and alignment perspectives. Our analysis of CLIP-based similarity scores revealed that ViT-B/32 offers richer semantic alignments compared to RN50, which tends to be more conservative. This suggests that transformer-based architectures like ViT-B/32 may better capture multimodal relationships, explaining TiQE's superior performance.However, some observations highlighted areas for improvement.The model overestimated the quality of images with lower MOS values, likely due to its emphasis on visually appealing features like color saturation. Additionally, the alignment-based predictions showed overestimation for images with lower ground-truth values, indicating a bias towards structural consistency over perceptual distortions.Future work could refine TiQE to better capture subtle perceptual distortions and improve its sensitivity to local quality degradations, potentially enhancing its performance on lower-quality images.

References

[1] Arjovsky, M., Chintala, S., Bottou, L.: Wasserstein GAN. In: Proceedings of the 34th International Conference on Machine Learning (ICML), pp. 214–223. PMLR (2017)

[2] Wang, P., Sun, W., Zhang, Z., Jia, J., Jiang, Y., Zhang, Z., Min, X., Zhai, G.: Large Multi-modality Model Assisted AI-Generated Image Quality Assessment. In: Proceedings of the 32nd ACM International Conference on Multimedia (MM '24), pp. 7803–7812. ACM (2024)

[3] Ye Q, Xu H, Ye J, Yan M, Hu A, Liu H, Qian Q, Zhang J, Huang F. mplug-owl2: Revolutionizing multi-modal large language model with modality collaboration. In-Proceedings of the ieee/cvf conference on computer vision and pattern recognition 2024 (pp. 13040-13051).

[4] Peng, F., Fu, H., Ming, A., Wang, C., Ma, H., He, S., Dou, Z., Chen, S.: AIGC Image Quality Assessment via Image-Prompt Correspondence. In: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) Workshops, pp. 6432–6441. IEEE (2024)

[5] Radford A, Kim JW, Hallacy C, Ramesh A, Goh G, Agarwal S, Sastry G, Askell A, Mishkin P, Clark J, Krueger G. Learning transferable visual models from natural language supervision. InInternational conference on machine learning 2021 Jul 1 (pp. 8748-8763). PmLR.

[6] Li J, Li D, Savarese S, Hoi S. Blip-2: Bootstrapping language-image pre-training with frozen image encoders and large language models. InInternational conference on machine learning 2023 Jul 3 (pp. 19730-19742). PMLR.

[7] Jia C, Yang Y, Xia Y, Chen YT, Parekh Z, Pham H, Le Q, Sung YH, Li Z, Duerig T. Scaling up visual and vision-language representation learning with noisy text supervision. InInternational conference on machine learning 2021 Jul 1 (pp. 4904-4916). PMLR.

[8] Oquab, M., Darcet, T., Moutakanni, T., Vo, H., Szafraniec, M., Khalidov, V., Fernandez, P., Haziza, D., Massa, F., El-Nouby, A., Assran, M., Ballas, N., Galuba, W., Howes, R., Huang, P., Li, S., Misra, I., Rabbat, M., Sharma, V., Synnaeve, G., Xu, H., Jegou, H., Mairal, J., Labatut, P., Joulin, A., Bojanowski, P.: DINOv2: Learning Robust Visual Features without Supervision. arXiv preprint arXiv:2304.07193 (2024)

[9] Abdul Samadh J, Gani MH, Hussein N, Khattak MU, Naseer MM, Shahbaz Khan F, Khan SH. Align your prompts: Test-time prompting with distribution alignment for zero-shot generalization. Advances in Neural Information Processing Systems. 2023 Dec 15;36:80396-413.

[10] Girdhar R, El-Nouby A, Liu Z, Singh M, Alwala KV, Joulin A, Misra I. Imagebind: One embedding space to bind them all. InProceedings of the IEEE/CVF conference on computer vision and pattern recognition 2023 (pp. 15180-15190).

[11] Yuan J, Yang F, Li J, Cao X, Che J, Lin J, Cao X. PKU-AIGIQA-4K: A perceptual quality assessment database for both text-to-image and image-to-image AI-generated images. arXiv preprint arXiv:2404.18409. 2024 Apr 29.

[12] Xia, J., He, L., Gao, F., Zhang, K., Li, L., Gao, X.: AI-Generated Image Quality Assessment Based on Task-Specific Prompt and Multi-Granularity Similarity. arXiv preprint arXiv:2411.16087 (2024)

[13] Li C, Zhang Z, Wu H, Sun W, Min X, Liu X, Zhai G, Lin W. Agiqa-3k: An open database for ai-generated image quality assessment. IEEE Transactions on Circuits and Systems for Video Technology. 2023 Sep 25;34(8):6833-46.

[14] Mittal, A., Moorthy, A.K., Bovik, A.C.: No-Reference Image Quality Assessment in the Spatial Domain. IEEE Transactions on Image Processing, 21(12), 4695–4708 (2012)

[15] Zhang, L., Zhang, L., Bovik, A.C.: A Feature-Enriched Completely Blind Image Quality Evaluator. IEEE Transactions on Image Processing, 24(8), 2579–2591 (2015)

[16] Mittal, A., Soundararajan, R., Bovik, A.C.: Making a "Completely Blind" Image Quality Analyzer. IEEE Signal Processing Letters, 20(3), 209–212 (2013)

[17] Zhang, W., Ma, K., Yan, J., Deng, D., Wang, Z.: Blind Image Quality Assessment Using a Deep Bilinear Convolutional Neural Network. IEEE Transactions on Circuits and Systems for Video Technology, 30(1), (2020)

[18] Su, S., Yan, Q., Zhu, Y., Zhang, C., Ge, X., Sun, J., Zhang, Y.: Blindly Assess Image Quality in the Wild Guided by a Self-Adaptive Hyper Network. In: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR). IEEE (2020)

[19] Cheon M, Yoon SJ, Kang B, Lee J. Perceptual image quality assessment with transformers. InProceedings of the IEEE/CVF conference on computer vision and pattern recognition 2021 (pp. 433-442).

[20] Ke J, Wang Q, Wang Y, Milanfar P, Yang F. Musiq: Multi-scale image quality transformer. InProceedings of the IEEE/CVF international conference on computer vision 2021 (pp. 5148-5157).
[21] Bosse, S., Maniry, D., Müller, K.R., Wiegand, T., Samek, W.: Deep Neural Networks for No-Reference and Full-Reference Image Quality Assessment. IEEE Transactions on Image Processing, 27(1), 206–219 (2018)
[22] Fardo FA, Conforto VH, De Oliveira FC, Rodrigues PS. A formal evaluation of PSNR as quality measurement parameter for image segmentation algorithms. arXiv preprint arXiv:1605.07116. 2016 May 23.
[23] Wang, Z., Bovik, A.C., Sheikh, H.R., Simoncelli, E.P.: Image quality assessment: from error visibility to structural similarity. IEEE Transactions on Image Processing, 13(4), 600–612 (2004)
[24] Topiwala P, Dai W, Pian J, Biondi K, Krovvidi A. VMAF and variants: towards a unified VQA. InApplications of Digital Image Processing XLIV 2021 Aug 1 (Vol. 11842, pp. 96-104). SPIE.
[25] Zhang R, Isola P, Efros AA, Shechtman E, Wang O. The unreasonable effectiveness of deep features as a perceptual metric. InProceedings of the IEEE conference on computer vision and pattern recognition 2018 (pp. 586-595).
[26] Ding, K., Ma, K., Wang, S., Simoncelli, E.: Image Quality Assessment: Unifying Structure and Texture Similarity. IEEE Transactions on Pattern Analysis and Machine Intelligence, pp. 1–1 (2020)
[27] Wang J, Chan KC, Loy CC. Exploring clip for assessing the look and feel of images. InProceedings of the AAAI conference on artificial intelligence 2023 Jun 26 (Vol. 37, No. 2, pp. 2555-2563).
[28] Barratt S, Sharma R. A note on the inception score. arXiv preprint arXiv:1801.01973. 2018 Jan 6.
[29] Heusel M, Ramsauer H, Unterthiner T, Nessler B, Hochreiter S. Gans trained by a two time-scale update rule converge to a local nash equilibrium. Advances in neural information processing systems. 2017;30.
[30] Huang Z, He Y, Yu J, Zhang F, Si C, Jiang Y, Zhang Y, Wu T, Jin Q, Chanpaisit N, Wang Y. Vbench: Comprehensive benchmark suite for video generative models. InProceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition 2024 (pp. 21807-21818).
[31] Kirstain Y, Polyak A, Singer U, Matiana S, Penna J, Levy O. Pick-a-pic: An open dataset of user preferences for text-to-image generation. Advances in neural information processing systems. 2023 Dec 15;36:36652-63.
[32] Golestaneh SA, Dadsetan S, Kitani KM. No-reference image quality assessment via transformers, relative ranking, and self-consistency. InProceedings of the IEEE/CVF winter conference on applications of computer vision 2022 (pp. 1220-1230).
[33] Yang S, Wu T, Shi S, Lao S, Gong Y, Cao M, Wang J, Yang Y. Maniqa: Multi-dimension attention network for no-reference image quality assessment. InProceedings of the IEEE/CVF conference on computer vision and pattern recognition 2022 (pp. 1191-1200).
[34] Cai W, Jiang J, Wang F, Tang J, Kim S, Huang J. A survey on mixture of experts in large language models. IEEE Transactions on Knowledge and Data Engineering. 2025 Mar 25.
[35] Xu J, Liu X, Wu Y, Tong Y, Li Q, Ding M, Tang J, Dong Y. Imagereward: Learning and evaluating human preferences for text-to-image generation. Advances in Neural Information Processing Systems. 2023 Dec 15;36:15903-35.